

Post Translational Modification localisation

Red = oxidation

Blue = deamidation

COL1A1 – (200-250)

Female Juvenile

SQMSYGYDEKSAGVSVPGPMGPSGPRGLPGPPGAPGPQGFQGPPEGEPEP
GASGPMGPRGPPGPPGKNGDDGEAGKPGRPGERGPPGPQGARGLPGTAGL
PGMKGHRGFSGLDGAKGDTGPAGPKGEPGSPGENGAPGQMGRGLPGERG

Female Adult

SQMSYGYDEKSAGVSVPGPMGPSGPRGLPGPPGAPGPQGFQGPPEGEPEP
GASGPMGPRGPPGPPGKNGDDGEAGKPGRPGERGPPGPQGARGLPGTAGL
PGMKGHRGFSGLDGAKGDTGPAGPKGEPGSPGENGAPGQMGRGLPGERG

COL1A1 – (950-1000)

Female Juvenile

ETGPAGRPGEVGP GPPGPAGEKGS PGADGPAGSPGTPGPQGIAGQRGVV
GLPGQ RGERGF PGLPGPSGEPGKQGPSGASGERGPPGPMGPPGLAGPPGE
SGREGSPGAEGSPGRDGAPGAKGDRGETGPAGPPGAPGAPGAPGPVGPAG

Female Adult

ETGPAGRPGEVGP GPPGPAGEKGS PGADGPAGSPGTPGPQGIAGQRGVV
GLPGQ RGERGF PGLPGPSGEPGKQGPSGASGERGPPGPMGPPGLAGPPGE
SGREGSPGAEGSPGRDGAPGAKGDRGETGPAGPPGAPGAPGAPGPVGPAG

COL1A1 – (1050-1100)

Female Juvenile

SGREGSPGAEGSPGRDGAPGAKGDRGETGPAGPPGAPGAPGAPGPVGPAG
KNGDRGETGPAGPAGPIGPAGARGPAGPQGPRGDKGETGEQGDRGIKGHR
GFSGLQ GPPGSPGSPGEQGPSGASGPAGPRGPPGSAGSPGKDGLNGLPGP

Female Adult

SGREGSPGAEGSPGRDGAPGAKGDRGETGPAGPPGAPGAPGAPGPVGPAG
KNGDRGETGPAGPAGPIGPAGARGPAGPQGPRGDKGETGEQGDRGIKGHR
GFSGLQ GPPGSPGSPGEQGPSGASGPAGPRGPPGSAGSPGKDGLNGLPGP

COL1A1 – (150-200)

Female Juvenile

EGPKGDGPQGPRGPVGP GPPQDGIPGQPGLPGPPGPPGPPGPPGLGGNFA
SQMSYGYDEKSAGVSVPGPMGPSGPRGLPGPPGAPGPQGFQGPPEGEPEP
GASGPMGPRGPPGPPGKNGDDGEAGKPGRPGERGPPGPQGARGLPGTAGL

Female Jumbo

EGPKGDGPQGPRGPVGP GPPQDGIPGQPGLPGPPGPPGPPGPPGLGGNFA
SQMSYGYDEKSAGVSVPGPMGPSGPRGLPGPPGAPGPQGFQGPPEGEPEP
GASGPMGPRGPPGPPGKNGDDGEAGKPGRPGERGPPGPQGARGLPGTAGL

COL1A1 – (1200-1250)

Female Juvenile

IGPPGPRGRTGDSGPAGPPGPPGPPGPPSGGYDFSFLPQPPQEKSQD
GGYYRADDANVVRDRDLEVDTTLSLSQQIENIRSPEGSRKNPARTCRD
LKMCHSDWKSGEYWIDPNQGCNLDAIKVYCNMETGQTCVFPQTQPSVPQKN

Female Adult

IGPPGPRGRTGDSGPAGPPGPPGPPGPPSGGYDFSFLPQPPQEKSQD
GGYYRADDANVVRDRDLEVDTTLSLSQQIENIRSPEGSRKNPARTCRD
LKMCHSDWKSGEYWIDPNQGCNLDAIKVYCNMETGQTCVFPQTQPSVPQKN

COL1A1 – (1000-1050)

Female Juvenile

KGERGYPGNIGPTGAAGAPGPHGSVGPAGKHGNRGEPGPAGSVGPVGAVG
PRGPSGPQGIRGDKGEPGDKGARGLPGLKGHNGLQGLPGLAGLHGDQGAP
GPVGPAGPRGPAGPSGPIGKDGRSGHPGPVGPAGVRGSQGSQGPAGPPGP

Female Adult

KGERGYPGNIGPTGAAGAPGPHGSVGPAGKHGNRGEPGPAGSVGPVGAVG
PRGPSGPQGIRGDKGEPGDKGARGLPGLKGHNGLQGLPGLAGLHGDQGAP
GPVGPAGPRGPAGPSGPIGKDGRSGHPGPVGPAGVRGSQGSQGPAGPPGP

Male Juvenile

KGERGYPGNIGPTGAAGAPGPHGSVGPAGKHGNRGEPGPAGSVGPVGAVG
PRGPSGPQGIRGDKGEPGDKGARGLPGLKGHNGLQGLPGLAGLHGDQGAP
GPVGPAGPRGPAGPSGPIGKDGRSGHPGPVGPAGVRGSQGSQGPAGPPGP

Male Adult

KGERGYPGNIGPTGAAGAPGPHGSVGPAGKHGNRGEPGPAGSVGPVGAVG
PRGPSGPQGIRGDKGEPGDKGARGLPGLKGHNGLQGLPGLAGLHGDQGAP
GPVGPAGPRGPAGPSGPIGKDGRSGHPGPVGPAGVRGSQGSQGPAGPPGP

COL1A2 – (1000-1050)

Female Juvenile

PRG**P**SG**PQ**GIRGDKGEPGDKGARGLPGL**KGH**NGL**QGL**PGLAGLHGD**Q**GAP
GPVGPAGPR**G**PAGPSGPIGKDGRSGH**P**GPVGPAGVRGSQGSQGPAGPPGP
PGPPGPPGVSGGGYDFGFEGGFYRADQPRSQPSLRPKDYEVDATLKSLNN

Female Adult

PRG**P**SG**PQ**GIRGDKGEPGDKGARGLPGL**KGH**NGL**QGL**PGLAGLHGD**Q**GAP
GPVGPAGPR**G**PAGPSGPIGKDGRSGH**P**GPVGPAGVRGSQGSQGPAGPPGP
PGPPGPPGVSGGGYDFGFEGGFYRADQPRSQPSLRPKDYEVDATLKSLNN

Male Juvenile

PRG**P**SG**PQ**GIRGDKGEPGDKGARGLPGL**KGH**NGL**QGL**PGLAGLHGD**Q**GAP
GPVGPAGPR**G**PAGPSGPIGKDGRSGH**P**GPVGPAGVRGSQGSQGPAGPPGP
PGPPGPPGVSGGGYDFGFEGGFYRADQPRSQPSLRPKDYEVDATLKSLNN

Male Adult

PRG**P**SG**PQ**GIRGDKGEPGDKGARGLPGL**KGH**NGL**QGL**PGLAGLHGD**Q**GAP
GPVGPAGPR**G**PAGPSGPIGKDGRSGH**P**GPVGPAGVRGSQGSQGPAGPPGP
PGPPGPPGVSGGGYDFGFEGGFYRADQPRSQPSLRPKDYEVDATLKSLNN

COL2A1 – (550-600)

Female Juvenile

AGPQGVG**P**SGAP**P**GEDGR**P**GP**PQ**GARG**Q**P**GV**M**GF**P**GP**K**G**ANG**E**P**G**KAG
EKGLAGAPGLRGL**P**GKDGETGAAG**PP**GPSGPAGERGE**Q**GAP**GP**SGFQGL**P**
GPPGPPGEGG**K**QGDQGI**P**GEAGAPGLVGPRGERGFPGERGSPGAQGLQGP

Female Adult

AGPQGVG**P**SGAP**P**GEDGR**P**GP**PQ**GARG**Q**P**GV**M**GF**P**GP**K**G**ANG**E**P**G**KAG
EKGLAGAPGLRGL**P**GKDGETGAAG**PP**GPSGPAGERGE**Q**GAP**GP**SGFQGL**P**
GPPGPPGEGG**K**QGDQGI**P**GEAGAPGLVGPRGERGFPGERGSPGAQGLQGP

Male Juvenile

AGPQGVG**P**SGAP**P**GEDGR**P**GP**PQ**GARG**Q**P**GV**M**GF**P**GP**K**G**ANG**E**P**G**KAG
EKGLAGAPGLRGL**P**GKDGETGAAG**PP**GPSGPAGERGE**Q**GAP**GP**SGFQGL**P**
GPPGPPGEGG**K**QGDQGI**P**GEAGAPGLVGPRGERGFPGERGSPGAQGLQGP

Male Adult

AGPQGVG**P**SGAP**P**GEDGR**P**GP**PQ**GARG**Q**P**GV**M**GF**P**GP**K**G**ANG**E**P**G**KAG
EKGLAGAPGLRGL**P**GKDGETGAAG**PP**GPSGPAGERGE**Q**GAP**GP**SGFQGL**P**
GPPGPPGEGG**K**QGDQGI**P**GEAGAPGLVGPRGERGFPGERGSPGAQGLQGP

COL2A1 – (550-600)

Female Juvenile

AGPQGKVGPSGAPGEDGRPGPPGPGARGQPGVMGFPGPKGANGEPGKAG
EKGLAGAPGLRGLPGKDGETGAAGPPGPSGPAGERGEQGAPGPSGFQGLP
GPPGPPGEGGKQGDQGIPGEAGAPGLVGPRGERGFPGERGSPGAQGLQGP

Female Adult

AGPQGKVGPSGAPGEDGRPGPPGPGARGQPGVMGFPGPKGANGEPGKAG
EKGLAGAPGLRGLPGKDGETGAAGPPGPSGPAGERGEQGAPGPSGFQGLP
GPPGPPGEGGKQGDQGIPGEAGAPGLVGPRGERGFPGERGSPGAQGLQGP

Male Juvenile

AGPQGKVGPSGAPGEDGRPGPPGPGARGQPGVMGFPGPKGANGEPGKAG
EKGLAGAPGLRGLPGKDGETGAAGPPGPSGPAGERGEQGAPGPSGFQGLP
GPPGPPGEGGKQGDQGIPGEAGAPGLVGPRGERGFPGERGSPGAQGLQGP

Male Adult

AGPQGKVGPSGAPGEDGRPGPPGPGARGQPGVMGFPGPKGANGEPGKAG
EKGLAGAPGLRGLPGKDGETGAAGPPGPSGPAGERGEQGAPGPSGFQGLP
GPPGPPGEGGKQGDQGIPGEAGAPGLVGPRGERGFPGERGSPGAQGLQGP

COL2A1 – (800-850)

Male Juvenile

SGAPGPQGPTGVTGPKGARGAQGPPGATGFPGAAGRVGPPGSNGNPGPAG
PPGPAGKDGPKGARGDTGAPGRAGDPGLQGPAGAPGEKGEPEGDDGPSGSD
GPPGPQGLAGQRGIVGLPGQRGERGFPLPGPSGEPGKQGAPGASGDRGP

Male Adult

SGAPGPQGPTGVTGPKGARGAQGPPGATGFPGAAGRVGPPGSNGNPGPAG
PPGPAGKDGPKGARGDTGAPGRAGDPGLQGPAGAPGEKGEPEGDDGPSGSD
GPPGPQGLAGQRGIVGLPGQRGERGFPLPGPSGEPGKQGAPGASGDRGP

COL2A1 – (1150-1200)

Female Juvenile

VGPSGKDGSNGIPGPIGPPGPRGRSGETGPAGPPGNPGPPGPPGPPGI
DMSAFAGLGQREKGPDPLOYMRADEADSTLRQHDEVDATLKSLNNQIES
IRSPDGSRKNPARTCQDLKLCHPEWKSGDYWIDPNQGCTLDAMKVFNCME

Female Adult

VGPSGKDGSNGIPGPIGPPGPRGRSGETGPAGPPGNPGPPGPPGPPGPI
DMSAFAGLGQREKGPDP LQYMRAD EADSTLRQH DVEVDATLKSLNNQIES
IRSPDGSRKNPARTCQDLKLCHPEWKSGDYWIDPNQGCTLDAMKVFCNME

Male Juvenile

VGPSGKDGSNGIPGPIGPPGPRGRSGETGPAGPPGNPGPPGPPGPPGPI
DMSAFAGLGQREKGPDP LQYMRAD EADSTLRQH DVEVDATLKSLNNQIES
IRSPDGSRKNPARTCQDLKLCHPEWKSGDYWIDPNQGCTLDAMKVFCNME

Male Adult

VGPSGKDGSNGIPGPIGPPGPRGRSGETGPAGPPGNPGPPGPPGPPGPI
DMSAFAGLGQREKGPDP LQYMRAD EADSTLRQH DVEVDATLKSLNNQIES
IRSPDGSRKNPARTCQDLKLCHPEWKSGDYWIDPNQGCTLDAMKVFCNME

COL2A1 – (1200-1250)

Female Juvenile

DMSAFAGLGQREKGPDP LQYMRAD EADSTLRQH DVEVDATLKSLNNQIES
IRSPDGSRKNPARTCQDLKLCHPEWKSGDYWIDPNQGCTLDAMKVFCNME
TGESCVYPNPATVPRKNWSSKSKEKKHIWFGETMNGGFHFSYGDGNLAP

Female Adult

DMSAFAGLGQREKGPDP LQYMRAD EADSTLRQH DVEVDATLKSLNNQIES
IRSPDGSRKNPARTCQDLKLCHPEWKSGDYWIDPNQGCTLDAMKVFCNME
TGESCVYPNPATVPRKNWSSKSKEKKHIWFGETMNGGFHFSYGDGNLAP

Male Juvenile

DMSAFAGLGQREKGPDP LQYMRAD EADSTLRQH DVEVDATLKSLNNQIES
IRSPDGSRKNPARTCQDLKLCHPEWKSGDYWIDPNQGCTLDAMKVFCNME
TGESCVYPNPATVPRKNWSSKSKEKKHIWFGETMNGGFHFSYGDGNLAP

Male Adult

DMSAFAGLGQREKGPDP LQYMRAD EADSTLRQH DVEVDATLKSLNNQIES
IRSPDGSRKNPARTCQDLKLCHPEWKSGDYWIDPNQGCTLDAMKVFCNME
TGESCVYPNPATVPRKNWSSKSKEKKHIWFGETMNGGFHFSYGDGNLAP

COL2A1 – (1200-1250)

Female Juvenile

DMSAFAGLGQREKGPDP LQYMRAD EADSTLRQH DVEVDATLKS LNNQIES
IRSPDGSRK NPARTCQDLKLCHPEWKSGDYWIDPNQGCTLDAMKVFCNME
TGESCVPNPATVPRKNWSSKSKEKKHIWFGETMNGGFHFSYGDGNLAP

Female Adult

DMSAFAGLGQREKGPDP LQYMRAD EADSTLRQH DVEVDATLKS LNNQIES
IRSPDGSRK NPARTCQDLKLCHPEWKSGDYWIDPNQGCTLDAMKVFCNME
TGESCVPNPATVPRKNWSSKSKEKKHIWFGETMNGGFHFSYGDGNLAP

Male Juvenile

DMSAFAGLGQREKGPDP LQYMRAD EADSTLRQH DVEVDATLKS LNNQIES
IRSPDGSRK NPARTCQDLKLCHPEWKSGDYWIDPNQGCTLDAMKVFCNME
TGESCVPNPATVPRKNWSSKSKEKKHIWFGETMNGGFHFSYGDGNLAP

Male Adult

DMSAFAGLGQREKGPDP LQYMRAD EADSTLRQH DVEVDATLKS LNNQIES
IRSPDGSRK NPARTCQDLKLCHPEWKSGDYWIDPNQGCTLDAMKVFCNME
TGESCVPNPATVPRKNWSSKSKEKKHIWFGETMNGGFHFSYGDGNLAP

COL2A1 – (1350-1400)

Female Juvenile

NTANVQMTFLRLLSTEGSQNITYHCKNSIAYLDEAAGNLKKALLIQGSND
VEMRAEGNSRFTYTALKDGCTKHTGKWGKTII EYRSQKTSRLPIVDIAPM
DIGGPDQEFGVDIGPVCFL

Female Adult

NTANVQMTFLRLLSTEGSQNITYHCKNSIAYLDEAAGNLKKALLIQGSND
VEMRAEGNSRFTYTALKDGCTKHTGKWGKTII EYRSQKTSRLPIVDIAPM
DIGGPDQEFGVDIGPVCFL

Male Juvenile

NTANVQMTFLRLLSTEGSQNITYHCKNSIAYLDEAAGNLKKALLIQGSND
VEMRAEGNSRFTYTALKDGCTKHTGKWGKTII EYRSQKTSRLPIVDIAPM
DIGGPDQEFGVDIGPVCFL

Male Adult

NTANVQMTFLRLLSTEGSQNITYHCKNSIAYLDEAAGNLKKALLIQGSND
VEMRAEGNSRFTYTALKDGCTKHTGKWGKTII EYRSQKTSRLPIVDIAPM
DIGGPDQEFGVDIGPVCFL

COL11A1 – (500-550)

Male Juvenile

MGPPGLQGPSGLPGDPGDRGPPGRPGLPGADGLPGPPGTMLMLPFRYGGD
GSKGPTISAQEAQAAILQQARIALRGPPGPMGLTGRPGPVGGPGSAGAK
GESGDPGPQGPRGVQGPPTGKPGKRGRPGADGGRGMPGEPGSKGDRGF

Male Adult

MGPPGLQGPSGLPGDPGDRGPPGRPGLPGADGLPGPPGTMLMLPFRYGGD
GSKGPTISAQEAQAAILQQARIALRGPPGPMGLTGRPGPVGGPGSAGAK
GESGDPGPQGPRGVQGPPTGKPGKRGRPGADGGRGMPGEPGSKGDRGF

COL11A1 – (950-1000)

Male Juvenile

KGTSGGDGPPPPGERGPQGPQGPVGFPGPKGPPGPAGKDGLPGHPGQRG
ETGFQGKTGPPGPGGVVGPQGPPTGETGPIIGERGHPTGPPGEQGLPGAA
GKEGAKGDPGPQGISGKDGPAGIRGFPGERGLPGAQGAAPGLKGGEGPQGP

Male Adult

KGTSGGDGPPPPGERGPQGPQGPVGFPGPKGPPGPAGKDGLPGHPGQRG
ETGFQGKTGPPGPGGVVGPQGPPTGETGPIIGERGHPTGPPGEQGLPGAA
GKEGAKGDPGPQGISGKDGPAGIRGFPGERGLPGAQGAAPGLKGGEGPQGP

COL11A1 – (1350-1400)

Male Juvenile

GPKGNPGPVGFPGDPPGPEGPAGQDGVGGDKGEDGDPGQPPGPGPSGE
AGPPGPPGKRGPPGASGSEGRQGEKGAKGEAGAEGPPGKTGPVGPQGPSPG
KPGPEGLRGIPGPVGEQGLPGAAGQDGPPLGPPGLPGLKGDPGSKGEK

Male Adult

GPKGNPGPVGFPGDPPGPEGPAGQDGVGGDKGEDGDPGQPPGPGPSGE
AGPPGPPGKRGPPGASGSEGRQGEKGAKGEAGAEGPPGKTGPVGPQGPSPG
KPGPEGLRGIPGPVGEQGLPGAAGQDGPPLGPPGLPGLKGDPGSKGEK

COL11A1 – (1400-1450)

Male Juvenile

AGPPGPPGKRGPPGASGSEGRQGEKGAKGEAGAEGPPGKTGPVGPQGPSPG
KPGPEGLRGIPGPVGEQGLPGAAGQDGPPLGPPGLPGLKGDPGSKGEK
GHPGLIGLIGPPGEQGEKGDRGLPGTQGSPPGAKDGGIPGPAGPIGPPGP

Male Adult

AGPPGPPGKRGPPGASGSEGRQGEKGAKGEAGAEGPPGKTGPVGPQGPSPG
KPGPEGLRGIPGPVGEQGLPGAAGQDGPPLGPPGLPGLKGDPGSKGEK
GHPGLIGLIGPPGEQGEKGDRGLPGTQGSPPGAKDGGIPGPAGPIGPPGP

SPP1 – (0-50)

Female Juvenile

MRLAVVCFCLFGLASCLPVKVAEFGSSEEKAHYSKHSDAVATWLKPDPSQ
KQNLLAPQNSVSSEETDDFKQETLPSNSNESHDMDDDDDDDDGDHAES

Female Adult

MRLAVVCFCLFGLASCLPVKVAEFGSSEEKAHYSKHSDAVATWLKPDPSQ
KQNLLAPQNSVSSEETDDFKQETLPSNSNESHDMDDDDDDDDGDHAES

Male Juvenile

MRLAVVCFCLFGLASCLPVKVAEFGSSEEKAHYSKHSDAVATWLKPDPSQ
KQNL LAPQNSVSSEETDDFKQETLPSNSNESHDMDDDDDDDDGDHAES

Male Adult

MRLAVVCFCLFGLASCLPVKVAEFGSSEEKAHYSKHSDAVATWLKPDPSQ
KQNL LAPQNSVSSEETDDFKQETLPSNSNESHDMDDDDDDDDGDHAES

SPP1 – (150-200)

Female Juvenile

EDSVNSDESDESHHSDESDESFTASTQADVLTPIPTVDVPDGRGDSLAY
GLRSKSRSPVSDQYDPDATDEDLTSRMKSQESDEAIKVIPVAQRLSVPS
DQDSNGKTSHESSQLDEPSVETHSLEQSKEYKQRASHESTEQSDAIDSAE

Female Adult

EDSVNSDESDESHHSDESDESFTASTQADVLTPIPTVDVPDGRGDSLAY
GLRSKSRSPVSDQYDPDATDEDLTSRMKSQESDEAIKVIPVAQRLSVPS
DQDSNGKTSHESSQLDEPSVETHSLEQSKEYKQRASHESTEQSDAIDSAE

Prothrombin – (350-400)

Female Juvenile

HDGDESIAGRTTDAEFHTFFDERTFGLGEADCGLRPLFEKKSLTDKTEKE
LLDSYIDGRIVEGWDAEKGIAPWQVMLFRKSPQELLCGASLISDRWVLT
AHCILYPPWDKNFTENDLLVRIGKHSRTRYERNVEKISMLEKIYIHPRYN

Female Adult

HDGDESIAGRTTDAEFHTFFDERTFGLGEADCGLRPLFEKKSLTDKTEKE
LLDSYIDGRIVEGWDAEKGIAPWQVMLFRKSPQELLCGASLISDRWVLT
AHCILYPPWDKNFTENDLLVRIGKHSRTRYERNVEKISMLEKIYIHPRYN

APOE – (200-250)

Female Juvenile

LRKMRKRLMRDADDLQKRLAVYKAGAQEGAERGVS AIRERLGPLVEQGRQ
RTANL GAGAAQPLRDRAQALSDRIRGRLEEVGNQARDRLEEVRQMEEV
SKMEEQTQQIRLQAEIFQARIKGWFEPLVEDMQRQWANLMEKIQASVATN

Female Adult

LRKMRKRLMRDADDLQKRLAVYKAGAQEGAERGVS AIRERLGPLVEQGRQ
RTANL GAGAAQPLRDRAQALSDRIRGRLEEVGNQARDRLEEVRQMEEV
SKMEEQTQQIRLQAEIFQARIKGWFEPLVEDMQRQWANLMEKIQASVATN

Male Juvenile

LRKMRKRLMRDADDLQKRLAVYKAGAQEGAERGVS AIRERLGPLVEQGRQ
RTANL GAGAAQPLRDRAQALSDRIRGRLEEVGNQARDRLEEVRQMEEV
SKMEEQTQQIRLQAEIFQARIKGWFEPLVEDMQRQWANLMEKIQASVATN

Male Adult

LRKMRKRLMRDADDLQKRLAVYKAGAQEGAERGVS AIRERLGPLVEQGRQ
RTANLGAGAAQPLRDRAQALSDRIRGRLEEVGNQARDRLEEVRREQMEEVR
SKMEEQTQQIRLQAEIFQARIKGWFEP LVEDMQRQWANLMEKIQASVATN

APOE – (250-300)

Female Juvenile

RTANLGAGAAQPLRDRAQALSDRIRGRLEEVGNQARDRLEEVRREQMEEVR
SKMEEQTQQIRLQAEIFQARIKGWFEP LVEDMQRQWANLMEKIQASVATN
SIASTTVPLENQ

Female Adult

RTANLGAGAAQPLRDRAQALSDRIRGRLEEVGNQARDRLEEVRREQMEEVR
SKMEEQTQQIRLQAEIFQARIKGWFEP LVEDMQRQWANLMEKIQASVATN
SIASTTVPLENQ

Male Juvenile

RTANLGAGAAQPLRDRAQALSDRIRGRLEEVGNQARDRLEEVRREQMEEVR
SKMEEQTQQIRLQAEIFQARIKGWFEP LVEDMQRQWANLMEKIQASVATN
SIASTTVPLENQ

Male Adult

RTANLGAGAAQPLRDRAQALSDRIRGRLEEVGNQARDRLEEVRREQMEEVR
SKMEEQTQQIRLQAEIFQARIKGWFEP LVEDMQRQWANLMEKIQASVATN
SIASTTVPLENQ

APOA1 – (200-250)

Female Juvenile

KNAKEMQRHLKVVAEEFRDRMRVNADALRAKFGLYSDQMRENLAQRLTEI
KNHPTLIEYHTKASDHLKTLGEKAKPALDDLGGGLMPVLEAWKAKIMSMI
DEAKKKLNA

Female Adult

KNAKEMQRHLKVVAEEFRDRMRVNADALRAKFGLYSDQMRENLAQRLTEI
KNHPTLIEYHTKASDHLKTLGEKAKPALDDLGGGLMPVLEAWKAKIMSMI
DEAKKKLNA

APOA2 – (50-100)

Female Juvenile

MKLLAMVALLVTICSLEGALVRRQAAETDVQTLFSQYLQSLTDYGKDLME
KAQPSEIQNQAKAYFQNAQERLTPFVQRTGTNLMDFLSRLMSPEEKPAPA
AK

Female Adult

MKLLAMVALLVTICSLEGALVRRQAAETDVQTLFSQYLQSLTDYGKDLME
KAQPSEIQNQAKAYFQNAQERLTPFVQRTGTNLMDFLSRLMSPEEKPAPA
AK

LUM – (100-150)

Female Juvenile

MYCDDLKLSVPMVPPGIKYLYLRNNQIDHIDEKAFENVTDLQWLILDHN
LLENSKIKGKVFSLKQKLKHLINYNLTESTVGPLPKSLQDLQLANNKIS
KLGSFDGLVNLTFIYLOHNQLKEEAVSASLKGLKSLEYLDLSFNQMSKLP

Female Adult

MYCDDLKLSVPMVPPGIKYLYLRNNQIDHIDEKAFENVTDLQWLILDHN
LLENSKIKGKVFSLKQKLKHLINYNLTESTVGPLPKSLQDLQLANNKIS
KLGSFDGLVNLTFIYLOHNQLKEEAVSASLKGLKSLEYLDLSFNQMSKLP

LUM – (150-200)

Male Juvenile

LLENSKIKGKVFSLKQKLKHLINYNLTESTVGPLPKSLQDLQLANNKIS
KLGSFDGLVNLTFIYLOHNQLKEEAVSASLKGLKSLEYLDLSFNQMSKLP
AGLPTSLTLYLDNNKITNIPDEYFNRTGLQYLRLSHNELADSGVPGNS

Male Adult

LLENSKIKGKVFSLKQKLKHLINYNLTESTVGPLPKSLQDLQLANNKIS
KLGSFDGLVNLTFIYLOHNQLKEEAVSASLKGLKSLEYLDLSFNQMSKLP
AGLPTSLTLYLDNNKITNIPDEYFNRTGLQYLRLSHNELADSGVPGNS

COMP – (600-650)

Female Juvenile

VLNQGMIEVQTMNSDPGLAVGYTAFNGVDFEGTFHVNTATDDDYAGFIFG
YQDSSSFYVVMWKQMEQTYWQANPFRAVAEPIQLKAVKSSTGPGEQLRN
ALWHTGDTASQVRLWWDPRNVGWKDKTSYRWFLQHRPQVGYIRVRFYEG

Female Adult

VLNQGMIEVQTMNSDPGLAVGYTAFNGVDFEGTFHVNTATDDDYAGFIFG
YQDSSSFYVVMWKQMEQTYWQANPFRAVAEPIQLKAVKSSTGPGEQLRN
ALWHTGDTASQVRLWWDPRNVGWKDKTSYRWFLQHRPQVGYIRVRFYEG

Male Juvenile

VLNQGMIEVQTMNSDPGLAVGYTAFNGVDFEGTFHVNTATDDDYAGFIFG
YQDSSSFYVVMWKQMEQTYWQANPFRAVAEPIQLKAVKSSTGPGEQLRN
ALWHTGDTASQVRLWWDPRNVGWKDKTSYRWFLQHRPQVGYIRVRFYEG

Male Adult

VLNQGMIEVQTMNSDPGLAVGYTAFNGVDFEGTFHVNTATDDDYAGFIFG
YQDSSSFYVVMWKQMEQTYWQANPFRAVAEPIQLKAVKSSTGPGEQLRN
ALWHTGDTASQVRLWWDPRNVGWKDKTSYRWFLQHRPQVGYIRVRFYEG

COMP – (500-550)

Female Juvenile

NSAQQDSHDGKGDACDDDDNDGVPSRDNCRLVPNPGQEDNDRDGVGD
ACQGDFDADKVIDKIDVCPENAEVLTDFRAFQTVVLDPEGDAQIDPNWV
VLNQGMEIVQTMNSDPGLAVGYTAFNGVDFEGTFHVNTATDDDYAGFIG

Female Adult

NSAQQDSHDGKGKGACDDDDNDGVPSRDNCRLVPNPGQEDNDRDGVGD
ACQGDFDADKVIDKIDVCPENAEVLTDFRAFQTVVLDPEGDAQIDPNWV
VLNQGMEIVQTMNSDPGLAVGYTAFNGVDFEGTFHVNTATDDDYAGFIG

Male Juvenile

NSAQQDSHDGKGKGACDDDDNDGVPSRDNCRLVPNPGQEDNDRDGVGD
ACQGDFDADKVIDKIDVCPENAEVLTDFRAFQTVVLDPEGDAQIDPNWV
VLNQGMEIVQTMNSDPGLAVGYTAFNGVDFEGTFHVNTATDDDYAGFIG

Male Adult

NSAQQDSHDGKGKGACDDDDNDGVPSRDNCRLVPNPGQEDNDRDGVGD
ACQGDFDADKVIDKIDVCPENAEVLTDFRAFQTVVLDPEGDAQIDPNWV
VLNQGMEIVQTMNSDPGLAVGYTAFNGVDFEGTFHVNTATDDDYAGFIG

ACAN – (600-650)

Female Juvenile

RTPCVGDKDSSPGVRTYGVPRSSSETYDVYCYVDKLEGEVFFATQMEQFTF
QEAQAFCAAQNATLASTGQLYAAWSQGLDKCYAGWLADGTLRYPIVNP
ACGGDKPGVRTVYLYPNQTLGPDPLSKHHAFCFRGVSVVPSPGGTPTSPS

Female Adult

RTPCVGDKDSSPGVRTYGVPRSSSETYDVYCYVDKLEGEVFFATQMEQFTF
QEAQAFCAAQNATLASTGQLYAAWSQGLDKCYAGWLADGTLRYPIVNP
ACGGDKPGVRTVYLYPNQTLGPDPLSKHHAFCFRGVSVVPSPGGTPTSPS